

Roger Baiges Trilla

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EDUCATION

- EPFL** | Lausanne, Switzerland **Sep. 2026 – Jun. 2028 (Expected)**
Master of Science in Computer Science, AI & Data Science Specialization
- Exploring model architectures and self-supervised learning for foundation models in multiplex tissue imaging at LTS4.
- Universitat Politècnica de Catalunya (UPC)** | Barcelona, Spain **Sep. 2022 – Jun. 2026**
Bachelor of Science in Artificial Intelligence (240 ECTS) GPA 9.19/10 · Top 5%
- 7 Honors Distinctions: Computer Vision, NLP & DL, RL & Unsupervised Learning, Statistics & Optimization.
- KU Leuven** | Leuven, Belgium **Sep. 2025 – Jan. 2026**
Erasmus+ Exchange Program in Artificial Intelligence
- Graduate coursework in Advanced Machine Learning, Uncertainty in AI, Privacy, and Evolutionary Computation.

RESEARCH EXPERIENCE

- Computer Vision Center (CVC)** | Researcher, LAMP Group | Barcelona, Spain **Feb. 2026 – Present**
- Demonstrated that model adaptation can break compatibility without erasing knowledge: frozen classifiers, prototype banks, and retrieval indexes fail while the underlying capability remains recoverable.
 - Proposed late-intermediate activation-space repairs optimized through frozen downstream layers: expressive variants restored over 90% of lost performance in CLIP and DINO; a single internal vector surpassed far larger output-space MLP/attention repairs.
 - Developed readout-aware null-space projections that restricted repairs to task-invisible directions, preserving adapted-task accuracy and extending the method to Qwen retrieval without rebuilding the index.

INDUSTRY EXPERIENCE

- Serimag** | AI Researcher, R&D | Barcelona, Spain **Feb. 2025 – Jul. 2025**
- Pioneered Serimag's first trainable extraction prototype for its hardest banking document type, previously handled manually.
 - Built the end-to-end data and fine-tuning stack for a state-of-the-art vision-language model using LoRA, target-only loss masking, custom output tokens, and memory-efficient training while keeping all sensitive data on premises.
 - Achieved nearly 80% field-level automation at the required 98% accuracy threshold by calibrating token probabilities and routing only uncertain fields to manual review.

AWARDS AND FELLOWSHIPS

- Won four AI hackathons:** Barça Innovation Hack (FC Barcelona; 1st prize, selected from 3,500+ applicants), AINA Hack (BSC), AI and Auditing Challenge, and HackUPC Bunge Challenge.
- Elena Maseras Fellowship** (Computer Vision Center), 2026.
- University Entrance Distinction** (Government of Catalonia), 2022. Top 0.7% in the common examination phase.

PROJECTS

- IpurdIAR: Tactical Football Intelligence** (*Barça Innovation Hack*) **Mar. 2026**
- Built an agentic multimodal system that autonomously selected tools and combined player-tracking data, inferred match events, and time-aligned commentary to answer complex questions about football tactics with evidence-grounded explanations.
- CatGPT: Catalan Language Model** (*Personal Project*) **2024**
- Trained and released a 111M-parameter Catalan GPT-style model from scratch, including a custom tokenizer and instruction-tuned variant, and created the 300M-token synthetic Patufet corpus to overcome Catalan data scarcity.
- LaIA: Multimodal Administrative Assistant** (*AINA Hack*) **Nov. 2024**
- Built a multimodal RAG assistant for navigating Catalan public services, combining dynamic web retrieval with document and image understanding, speech interaction, and automated form completion.
- Other Projects & Hackathons**
- Built AI systems across 10+ hackathons and personal projects, releasing code and models on GitHub and Hugging Face.

SKILLS

Programming: Python, C/C++, SQL, R, PDDL.
Libraries & Frameworks: PyTorch, Transformers, TensorFlow, NumPy, Pandas, scikit-learn, OpenCV, Spark.
Engineering Tools: Git, Docker, Linux, CUDA.
Expertise: Foundation Models, Vision-Language Models, AI Agents & Agentic Workflows, Continual Learning, Mechanistic Interpretability, PEFT, Document AI, Uncertainty Estimation.
Languages: Catalan (Native), Spanish (Native), English (C1 Cambridge Advanced).